Farkle Implementation Part Two, Fiona Callahan

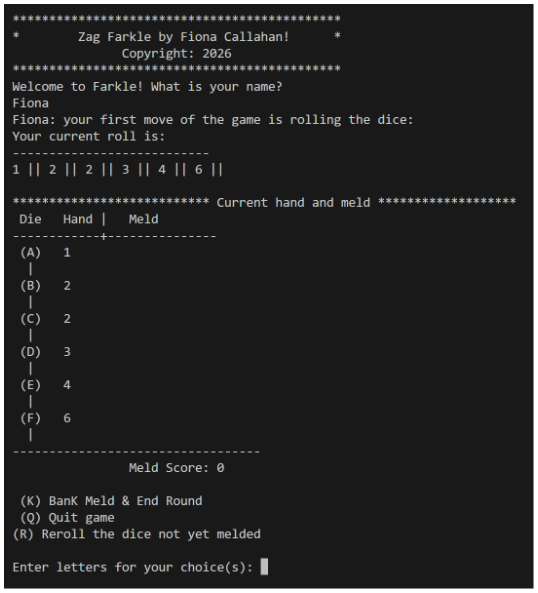
Goal Of My Program

- My program implementation, “Farkle”, models the real game that includes elements such as dice, the hand, scorecards, and melds. It allows single-players to log on and play the game. The user can roll their dice, and decide to move certain dice into the meld. For the dice that they keep in their hand, they are able to reroll. The new feature added in this program is the ability to detect a HotHand, however, the user must beware because a Farkle is still possible. If the user decides to quit, they do not gain their melded points, and instead, just are returned their total score. The main goal of the program is to make it user friendly, and make sure there are clear steps forward at every point.

General Design of the Program

- The design I chose for this program includes four different classes: Scorecard, PlayGame, HandOfDie, and Die. I chose this design because it blocked each set of functions off by way of similar instructions. I followed Java’s OOP style of programming. The Farkle class is what kicks off the program by creating a hand before it sends the user to “PlayGame”. PlayGame is what I see as the main, as it includes the method, checkFarkle, which leads the user through the roll. The Die, and the HandOfDie class go hand in hand because an object, hand, is full of the die objects. I oriented my program around ArrayLists, as I found this easiest to access variables and keep changing its length and values. The user is always shown different options for steps forward, and prompted when appropriate. The meld score is displayed at the bottom of each menu, and the total score is displayed at the end when the user is done with the game.

Program’s Starting Screen



Major Design/Programming issues

- The main design program that I found while coding Homework 2 was making sure that scores were updated at all the right spots. Due to implementing reroll, and quit game, there had to be some repeating actions of checking Farkle, and checking meld scores. I needed to figure out where to separate all the actions into. I also needed to stay on top of updating meld where appropriate, and making sure that the hand was rerolling the dice that were necessary.
- I found this assignment easier than the last, as the majority of setup was already done. However, I did find several inconsistencies that needed to be fixed with the new features added.

What would I have done differently

- I have enjoyed coming back to this assignment, and always finding ways to improve it. I believe that my implementation this time around is easier to follow, and it helps that it provides the user with more options each turn. Next time, I still would like to redesign the checkFarkle method. It has a lot of repetitive actions involved in it that I need to separate out. There could be a separate method of checking a farkle, and a separate method for checking the score each round. Especially if we will continue building on top, I never want to feel lost in my own program. I look forward to finding new ways to improve my user interface, as well. By implementing the name feature, next time, I want to be able to include more than one player. This would be pretty easy because I could create a Player object that holds their name, score, hand, etc.